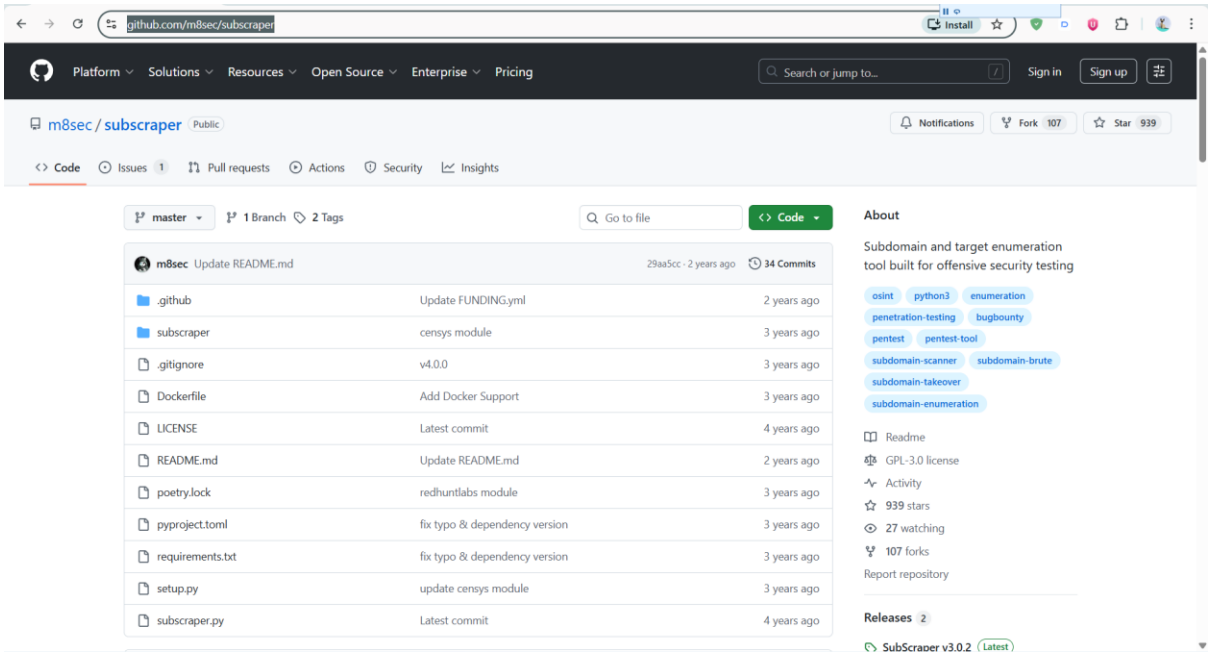
SUB-DOMAIN ENUMERATION TOOL "SUBSCRAPER" INSTALLATION AND USAGE IN KALI LINUX

Github url: *https://github.com/m8sec/subscraper*



Installation commands

Python

The following can be used to install SubScraper on Windows, Linux, & MacOS:

- ✓ git clone https://github.com/m8sec/subscraper
- ✓ cd subscraper
- ✓ pip3 install -r requirements.txt

Modules

A full list of modules that are used in performing the scanning.

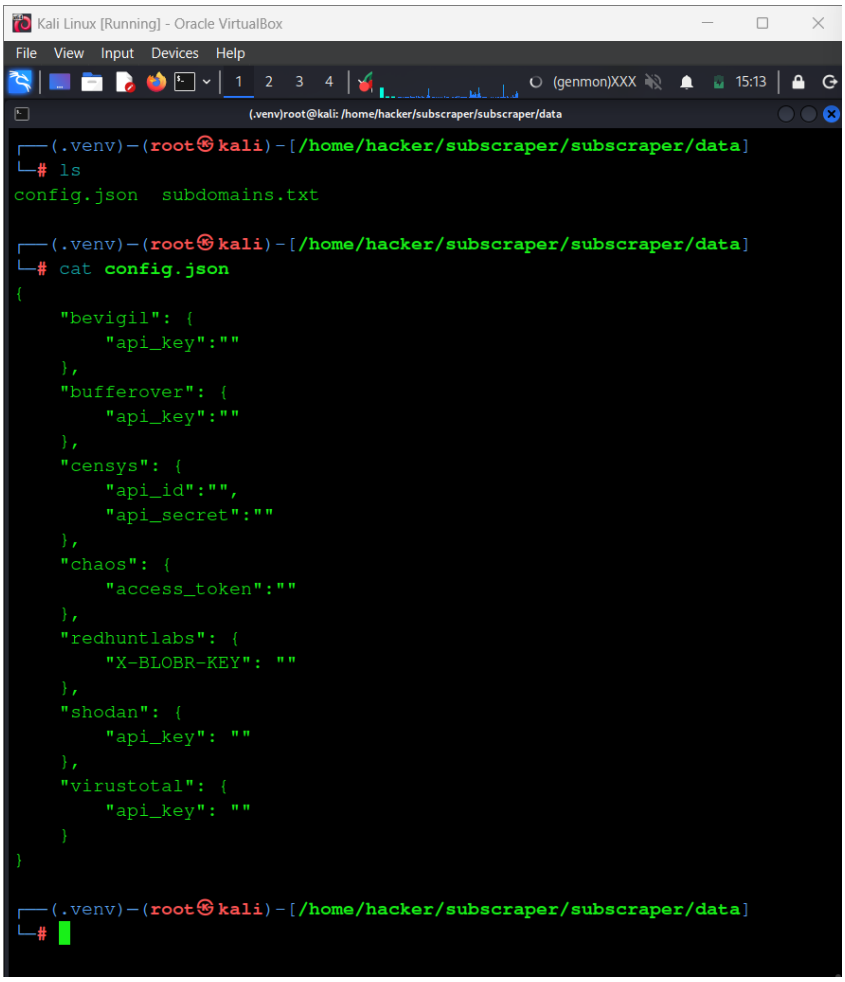
bevigil	- BeVigil OSINT API for scraping mobile application for subdomains (API Key Req)
crt.sh	- Subdomains enumeration using cert.sh.
virustotal	- Lookup subdomain on VirusTotal (API Key Req)
dnsrepo	- Parse dnsrepo.noc.org without an API key - 150 result limit
chaos	- Project Discovery's Chaos (API Key Req)

certspotter	- Use Certspotter API to collect subdomains
bufferover	- Query Bufferover.run API (API Key Req)
alienvault	- Find subdomains using AlienVault OTX
redhuntlabs	- RedHunt Labs recon API (API Key Req)
archive	- Use archive.org to find subdomains.
dnsdumpster	- Use DNS dumpster to enumerate subdomains.
censys.io	- Gather subdomains through censys.io SSL cert Lookups. (API Key Req)
shodan	- Get subdomains with Shodan (API Key Req)

Updates:

Use the configuration file at **~/subscraper/subscraper/data/config.json** to store **API key**s for easy reuse.

If updating to a newer version after v4.0.0, use the -update argument to pull a new copy of the config file and ensure compatibility - *Note: This will remove any existing key entries.*



Usage

Command Line Args

```
SubScraper Options:
  -debug              Enable debug logging
  -update             Update config file (Will remove existing entries)
  -config CONFIG      Override default config location
  -silent             Show subdomains only in output
  -threads THREADS, -T THREADS    Max threads for enumeration (65*).
  -t TIMEOUT                      set connection timeouts (3*)
  -d DOMAIN, --domain DOMAIN      Target domain input (domain, .txt, STDIN, etc.

Module Options:
  -ls                 List SubScraper enumeration modules.
  -m MODULES          Execute module(s) by name or group (all*).
  -module-only        Execute modules only not brute force

Bruteforce Options:
  -w WORDLIST         Custom wordlist for DNS brute force.
  -ns NS              Comma separated nameservers to use

Enumeration Options:
  -r, -resolve        Resolve IP address for each subdomain identified.
  -c, -cname          Perform CNAME lookup for subdomain takeover checks
  -http               Probe for active HTTP services.
  -http-port HTTP_PORT  HTTP ports to check, comma separated (80,443*)

Output Options:
  -nc, -no-color      Disable color output
  -active             Only report active subdomains with resolved IP
  -csv                Create CSV output report
  -o REPORT           Output file
```

Example Inputs

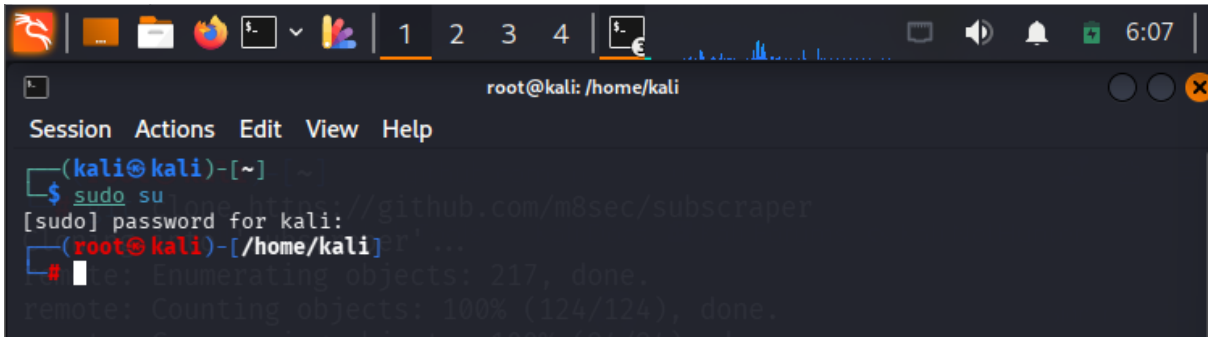
```
# python3 subscraper.py -d example.com -resolve -http -module-only

# python3 subscraper.py -d example.com -cname -m none -o sub_report.csv -csv

# cat domains.txt | python3 subscraper.py -active -silent
```

Steps to Install & work on it

Step 1: Get into root terminal or root previleges

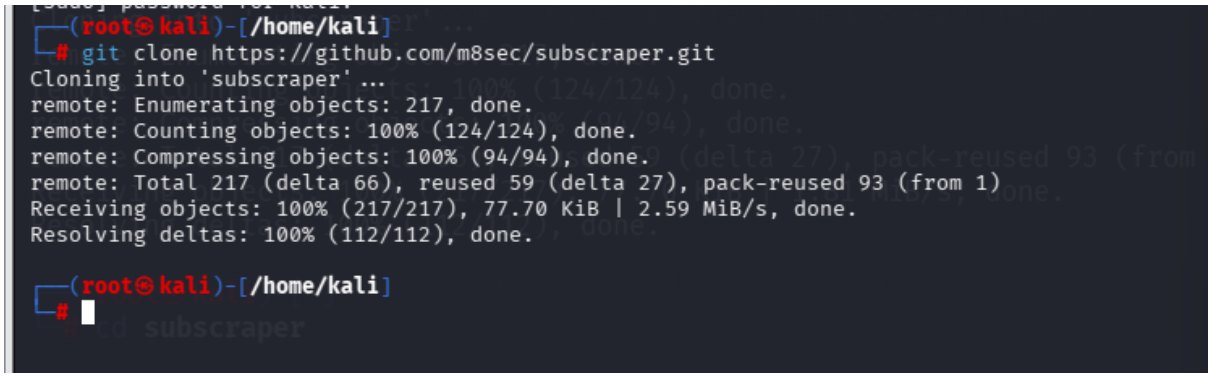
$ Sudo su
$ Enter password

Now you are having root previleges.



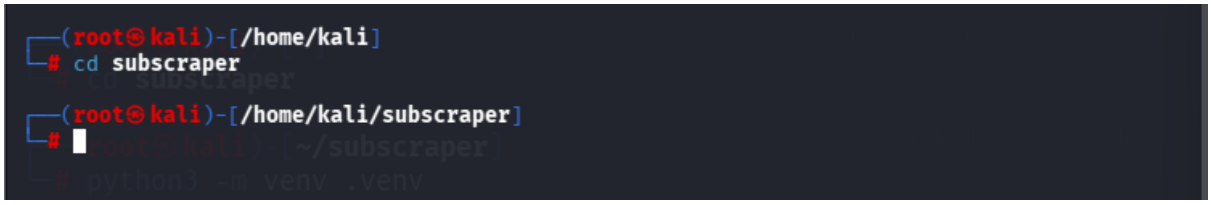
Step 2: Clone the github repository or paste the below command as-it-is.

 # git clone https://github.com/m8sec/subscraper.git



Step 3: Move into subscraper directory

cd subscraper



Step 4: Create a virtual environment inorder to not face any errors or issues, use below command.

python3 -m venv [virtual environment name of your intrest]

Ex: *# python3 -m venv Praveen*

```
┌──(root㉿kali)-[/home/kali/subscraper]
└─# python3 -m venv praveen

┌──(root㉿kali)-[/home/kali/subscraper]
└─#
```

Step 5: Activate the virtual environment, use the below command.

source [your virtual environment name used in previous step]/bin/activate

Ex: *#source praveen/bin/activate*

```
┌──(root㉿kali)-[/home/kali/subscraper]
└─# source praveen/bin/activate

┌──(praveen)─(root㉿kali)-[/home/kali/subscraper]
└─#
```

Step 6: Install requirements that are mentioned in the requirements.txt file.

 # pip3 install -r requirements.txt

```
┌──(praveen)─(root㉿kali)-[/home/kali/subscraper]
└─# pip3 install -r requirements.txt
Collecting censys≥2.2.6 (from -r requirements.txt (line 1))
  Using cached censys-2.2.19-py3-none-any.whl.metadata (7.7 kB)
Collecting taser≥0.4.0 (from -r requirements.txt (line 2))
  Using cached taser-0.4.4-py3-none-any.whl.metadata (2.5 kB)
Collecting ipparser≥1.0.1 (from -r requirements.txt (line 3))
  Using cached ipparser-1.0.2-py3-none-any.whl.metadata (2.5 kB)
Collecting argcomplete<4.0.0,≥2.0.0 (from censys≥2.2.6→-r requirements.txt (line 1))
```

Step 7: Perform scan. For better results use flags mentioned above in the "Usage" section.

Use command*:* *# python3 subscraper.py -d example.com*

```
┌──(praveen)─(root㉿kali)-[/home/kali/subscraper]
└─#  python3 subscraper.py -d tesla.com

 v4.0.2
 @m8sec

[dns-brute] www.tesla.com
[dns-brute] link.tesla.com
[dns-brute] sso.tesla.com
[dns-brute] people.tesla.com
[dns-brute] pay.tesla.com
[dns-brute] partners.tesla.com
[dns-brute] shop.tesla.com
[dns-brute] apps.tesla.com
[dns-brute] mobile.tesla.com
[dns-brute] marketing.tesla.com
[dns-brute] share.tesla.com
[dnsdumpster] accounts.tesla.com
[dnsdumpster] akamai-apigateway-vehicleextinfogw-prdsvc-st.tesla.com
[dnsdumpster] apac-cppm.tesla.com
[dnsdumpster] ams13-gpgw1.tesla.com
[dnsdumpster] assets-ir.tesla.com
[dnsdumpster] auth.tesla.com
[dnsdumpster] autodiscover.tesla.com
```

Resolve flag:

```
┌──(praveen)-(root㉿kali)-[/home/kali/subscraper]
└─#  python3 subscraper.py -d tesla.com -resolve

/home/kali/subscraper/subscraper/reporter.py:29: SyntaxWarning: invalid escape sequence '\.
'
  self.sub_regex = ".*[a-zA-Z0-9]([-a-zA-Z0-9]{0,61}[a-zA-Z0-9])?\.([a-zA-Z0-9]{1,2}([-a-zA
-Z0-9]{0,252}[a-zA-Z0-9])?)\.([a-zA-Z]{2,63})"
/home/kali/subscraper/subscraper/cli.py:13: SyntaxWarning: invalid escape sequence '\_'
  \__ \ || | '_ \__ \ V _| '_/ _` | '_ V -_) '_|
   /  __|   _| |__/  __| __ _ _ __ _ _ __  __ _ _
   \__ \ || | '_ \__ \ V _| '_/ _` | '_ V -_) '_|
   |___/\_,_|_.__/___/\__|_| \__,_| .__/\___|_| v4.0.2
                                   |_|            @m8sec

[dns-brute] www.tesla.com [104.120.132.60]
[dns-brute] link.tesla.com [104.120.132.60]
[dns-brute] sso.tesla.com [104.120.132.60]
[dns-brute] people.tesla.com [104.120.132.60]
[dns-brute] partners.tesla.com [104.120.132.60]
[dns-brute] pay.tesla.com [104.120.132.60]
```

http flag:

```
┌──(praveen)-(root㉿kali)-[/home/kali/subscraper]
└─#  python3 subscraper.py -d tesla.com -http

   /  __|   _| |__/  __| __ _ _ __ _ _ __  __ _ _
   \__ \ || | '_ \__ \ V _| '_/ _` | '_ V -_) '_|
   |___/\_,_|_.__/___/\__|_| \__,_| .__/\___|_| v4.0.2
                                   |_|            @m8sec

[dns-brute] pay.tesla.com [80/HTTP - Access Denied (403)] [443/HTTPS - Access Denied (403)]
[dns-brute] partners.tesla.com [80/HTTP - Access Denied (403)] [443/HTTPS - Access Denied (403)]
[dns-brute] www.tesla.com [80/HTTP - Access Denied (403)] [443/HTTPS - Access Denied (403)]
[dns-brute] shop.tesla.com [80/HTTP - Access Denied (403)] [443/HTTPS - Access Denied (403)]
[dns-brute] mobile.tesla.com [80/HTTP - Access Denied (403)] [443/HTTPS - Access Denied (403)]
[dns-brute] people.tesla.com [80/HTTP - Access Denied (403)] [443/HTTPS - Access Denied (403)]
[dns-brute] apps.tesla.com [80/HTTP - Access Denied (403)] [443/HTTPS - Access Denied (403)]
[dns-brute] link.tesla.com [80/HTTP - Access Denied (403)] [443/HTTPS - Access Denied (403)]
[dnsdumpster] assets-ir.tesla.com [80/HTTP - Access Denied (403)] [443/HTTPS - Access Denied (403)]
[dns-brute] sso.tesla.com [80/HTTP - Access Denied (403)] [443/HTTPS - N/A (404)]
[dnsdumpster] billing.tesla.com [80/HTTP - Access Denied (403)] [443/HTTPS - Access Denied (403)]
[dnsdumpster] bettertime.tesla.com [80/HTTP - Access Denied (403)] [443/HTTPS - Access Denied (403)]
[dnsdumpster] autodiscover.tesla.com [80/HTTP - Outlook (200)]
[dnsdumpster] bettertime-stage.tesla.com [80/HTTP - Access Denied (403)] [443/HTTPS - Access Denied (403)]
[dnsdumpster] cdn-design.tesla.com [80/HTTP - Access Denied (403)] [443/HTTPS - Access Denied (403)]
[dnsdumpster] auth.tesla.com [80/HTTP - Service Unavailable - Fai (503)..] [443/HTTPS - Access Denied (403)]
[dnsdumpster] akamai-apigateway-vehicleextinfogw-prdsvc-st.tesla.com [80/HTTP - Access Denied (403)] [443/HTTP
S - Access Denied (403)]
[dnsdumpster] media-server-qr.prd.america.vn.cloud.tesla.com [443/HTTPS - Tesla QRCode Service (200)]
```

```
[dnsrepo] uicrashtracer.tesla.com [104.120.132.60] [80/HTTP - Service Un
available - Fai (401)..] [443/HTTPS - Access Denied (401)]
[dnsrepo] external-automation.tesla.com [209.133.79.52]
[dnsrepo] mta2.emails.tesla.com [13.111.88.1]
[*] Identified 148 subdomain(s) in 0:00:20.980594.
[*] Subdomains written to ./sub_report.txt.
```

```
[crt.sh] vpn2.tesla.com [8.47.24.215]
[crt.sh] cnvpn.tesla.com [103.222.41.215]
[crt.sh] xmail.tesla.com [204.74.99.100] [80/HTTP - Error (200)]
[crt.sh] events.tesla.com [13.111.47.195]
[crt.sh] click.emails.tesla.com [13.111.48.179]
[crt.sh] mfamobile-dev.tesla.com [205.234.27.209]
^C^C[*] Identified 580 subdomain(s) in 0:01:09.935795.
[*] Subdomains written to ./sub_report.txt.
```